MMAUG

AI & DEVOPS BOOTCAMP

PROJECT 02

MMAUG High School Secure Login Portal

Request for Proposal

Customer	MMAUG High School
Location	Sliema, Malta
RFP Reference	MMAUG-AIDB-RFP-002
Document	Participant Copy
Status	Training Simulation

RFP Information	Details
Project	MMAUG High School Secure Login Portal
Customer	MMAUG High School
Location	Sliema, Malta
RFP Reference	MMAUG-AIDB-RFP-002
Project Type	Web Application / Identity / Security / DevOps
Audience	Bootcamp participant solution teams

1. Purpose of This RFP

This Request for Proposal is a simulated customer procurement document for the MMAUG AI & DevOps Bootcamp. Your team should respond as a professional engineering or consulting organisation. The customer has intentionally described business outcomes, constraints and acceptance expectations without prescribing a complete technical architecture.

Your task
Interpret the requirements, identify ambiguities, ask appropriate clarifying questions, propose and justify a solution, build a working MVP, and demonstrate how the solution can be securely delivered and operated using modern DevOps practices. The use of Agentic AI Coding assistants such as Claude Code, OpenAI Codex and GitHub Copilot for this project is allowed.

2. Customer Background

MMAUG High School is a fictitious secondary school serving approximately 1,200 students and 150 staff members. The school currently uses several disconnected systems and wants a central web portal through which students, teachers and administrators can securely access school services.

3. Business Problem

- Existing access controls are limited and inconsistent.
- User administration is largely manual.
- The school has limited visibility into sign-in activity and access attempts.
- Different user types require different information and privileges.
- The solution should establish a secure foundation for future school services.

4. Project Objectives

- Provide a secure central sign-in experience.
- Apply least-privilege role-based access.
- Provide role-appropriate dashboards and protected APIs.
- Enable controlled account administration and recovery.
- Provide usable audit and security telemetry.
- Deploy through repeatable DevOps practices rather than manual server changes.

5. Functional Requirements

- Support student, teacher and administrator user types.
- Provide secure sign-in, sign-out and account/password recovery.
- Present a dashboard appropriate to the authenticated role.
- Allow authorised administrators to view users, activate/deactivate accounts and assign permitted roles.

- Record relevant login history and audit events.
- Use appropriate session management and secure error handling.
- Prevent users from viewing resources outside their authorised role.
- Provide a protected API layer for application data.

6. Decision and Scope Boundaries

- Student information must not be exposed to other students without an explicit authorised use case.
- Administrative capabilities must not be accessible simply by altering a client-side value or URL.
- The application must not store passwords in plain text.

7. Security and Privacy Requirements

- Strong authentication and MFA where appropriate.
- Least privilege and server-side authorisation.
- Protection against credential attacks and repeated failed sign-ins.
- Secure session/token handling.
- Protection against common web application risks.
- API authorisation, encryption and secret management.
- Audit logging and security monitoring.
- Separation of privileged administration from standard user functions.

8. Responsible AI Requirements

- This project is primarily identity/security focused. If AI is added as a stretch feature, it must not weaken access controls or expose student information.

9. DevOps / DevSecOps Requirements

- All code in source control.
- Pipeline sequence should include build, tests, security validation, deployment and verification.
- Separate development, test and production environments.
- No manual copying of production application files.
- Environment-specific configuration and secrets managed outside source code.
- Deployment traceability and rollback capability.

10. MVP Demonstration Requirements

- Five student accounts, two teacher accounts and one administrator.
- Successful and failed authentication scenarios.
- Role-specific dashboards.
- Attempted unauthorised access and correct denial.
- Account recovery/reset workflow.
- Audit/logging capability.
- CI/CD deployment.
- A scale discussion for at least 1,500 registered accounts.

Response Requirements

Your proposal should be written for both business and technical stakeholders. At minimum, include the following areas where they are relevant to the project:

- Executive summary and understanding of the business problem.
- Functional and non-functional requirements identified from the RFP.
- Assumptions, dependencies, exclusions and clarifying questions.
- Proposed solution architecture and data flow.
- Application, API, data and integration design.
- AI, machine-learning or agent design where applicable.
- Identity, security, privacy and Responsible AI controls.
- DevOps/DevSecOps, CI/CD and Infrastructure-as-Code approach.
- Development, test/UAT and production environment strategy.
- Testing, monitoring, logging, rollback and operational support approach.
- Implementation phases, risks and mitigations.
- Working MVP demonstration and supporting technical documentation.

Technology choices must be justified. Microsoft Azure services may be proposed, but the RFP does not require a specific service merely because it exists.

Delivery Phases

Phase	Customer expectation
1 - Discovery & Requirements	Analyse the RFP, identify requirements and assumptions, and submit clarifying questions.
2 - Solution Design	Produce architecture, security model, data flow, AI approach where relevant, and DevOps design.
3 - Build & Integration	Implement iteratively using source control, automated testing and repeatable infrastructure/deployment practices.
4 - Validation & Deployment	Perform functional, integration and security validation before deployment.
5 - Demonstration & Handover	Demonstrate the solution, explain limitations and operational considerations, and hand over documentation.

General Evaluation Criteria

Evaluation area	Weight
Understanding of the business problem and requirements	10%
Solution architecture and technical design	15%
Functional implementation	15%
AI/ML/agent design where applicable	15%
Security, privacy and Responsible AI	15%
DevOps, CI/CD and Infrastructure as Code	15%
Testing, monitoring and reliability	10%
Documentation and customer presentation	5%

A sophisticated implementation that misses the business requirement is not a successful response. Equally, a functioning application deployed manually without appropriate security, testing and operational controls is incomplete.

Submission Reminder

Customer expectation
Your response should demonstrate the complete engineering lifecycle: **Understand -> Design -> Build -> Secure -> Test -> Deploy -> Monitor -> Improve.** Do not assume requirements that are not stated; document your assumptions and raise clarifying questions.